

Requirements Traceability Matrix

Student: Sujay Hegde | SRN: PES1UG24CS478

Project: BPS #47 - Domain & SSL Certificate Expiry Alert System

This matrix traces the seven baseline requirements to the use cases, proposed BPS #47 modules, work packages, and verification procedures. The coffee-kiosk diagram in folder 2 is a separate handout exercise and does not implement these requirements. Test coverage below is a plan; it does not establish that a deployed application meets a requirement.

Requirement	Summary / priority	Use cases	Proposed modules	WBS package	Verification IDs	Current evidence
FR-001	Daily TLS checks; alerts at 30, 15, 3 days and for expired certificates / High	UC-003 Scan; UC-004 Handle Alert	Scheduler, TLS Adapter, Alert Engine	WP-3.1, WP-3.3	TC-001, TC-002	Requirements and Jira BB-1; starter policy tests only
FR-002	Daily WHOIS checks; alerts at 45, 30, 7 days / High	UC-003; UC-004	Scheduler, WHOIS Adapter, Alert Engine	WP-3.2, WP-3.3	TC-003, TC-004	Requirements and Jira BB-2; injected lookup/retry tests only
FR-003	Acknowledge alerts; escalate high-priority alerts after 48h / Medium	UC-004; UC-005 Acknowledge; UC-006 Escalate	Alert Engine, Notification Adapter, Auth	WP-3.3, WP-3.4	TC-005, TC-006	Flow specification and Jira BB-3; starter state/timer tests only
FR-004	Authorized domain/ endpoint add, edit, remove / High	UC-001 Manage; UC-002 Authenticate	Registry, Auth, Dashboard	WP-2.1, WP-2.2	TC-007, TC-008	Jira BB-4; starter registry/ role tests only
FR-005	Monthly all-asset summary emailed to stakeholders / Low	UC-007 Monthly Report	Reporting, Notification Adapter, Scheduler	WP-3.5	TC-009, TC-010	Jira reporting work items; starter report fixture only
NFR-001	1,000 assets under 180s; HTTPS and role controls / High	All protected user operations; UC-003	Bounded-concurrency Scanner, Auth, HTTPS Gateway	WP-2.1, WP-4.1	TC-011, TC-012	Benchmark and deployed HTTPS verification pending
NFR-002	99.9% monthly availability; bounded backoff retries / High	UC-003; system-wide	Lookup Adapters, Monitoring, Persistence	WP-3.2, WP-4.2	TC-004, TC-013, TC-014	Starter retry/isolation tests only; availability measurement pending

Verification procedures

ID	Procedure	Expected result	Level / status
TC-001	Supply TLS expiry fixtures at 30, 15, 3, 31, and 0 days; repeat a scan	Alerts at specified thresholds and expiry; repeated threshold scan creates no duplicate	Unit fixtures implemented; live TLS integration pending
TC-002	Run the daily job against a controlled valid and expired TLS endpoint	Read certificate expiry, record an expired-certificate warning, and preserve validation errors	Integration / planned
TC-003	Supply WHOIS expiry fixtures at 45, 30, 7, and 46 days	Alerts at 45, 30, 7; no threshold alert at 46	Unit fixtures implemented; live WHOIS integration pending
TC-004	Fail a lookup transiently, then persistently	At most three retries after the initial attempt, increasing delays; persistent error recorded	Unit fixtures implemented
TC-005	Acknowledge an open alert before 48h; then run escalation	Actor/time recorded; no pending escalation is sent	Unit fixture implemented
TC-006	Check high-priority alert at 47h59m and 48h; check low-priority alert at 48h	Escalate once at/after 48h only for the unacknowledged high-priority alert	Unit fixture implemented
TC-007	Add, edit, and remove an asset; perform the next scan	Registry reflects edits; removed asset is not scanned	Unit fixture implemented; dashboard integration pending
TC-008	Attempt mutation as Security Officer and an unknown role	Both denied; authorized SysAdmin succeeds	Unit fixture implemented; session security pending
TC-009	Produce report after mixed successful and failed scans	Every current asset appears; unknown/failed results are explicit	Unit fixture implemented
TC-010	Exercise first-of-month schedule and stakeholder mail delivery	One report per month; delivery records and retry behavior retained	Integration / planned
TC-011	Scan 1,000 assets using a recorded environment and latency/failure profile	Total elapsed time below 180 seconds including timeout records	Performance / planned

ID	Procedure	Expected result	Level / status
TC-012	Visit deployed HTTPS dashboard; test unauthenticated access and role changes	HTTPS enforced; protected requests denied without valid authorization	Security / planned
TC-013	Fail one asset while another succeeds	Failed asset logged; other asset still scanned; worker stays alive	Unit fixture implemented
TC-014	Measure service health throughout one calendar month	Available service minutes / monitored minutes >= 99.9%	Operational / planned

Defect traceability

Captured Jira defect	Requirement	Planned regression
BB-1: Missing 15-day SSL alert	FR-001	TC-001, TC-002
BB-2: WHOIS failure without retry	FR-002, NFR-002	TC-004, TC-013
BB-3: Missing 48-hour escalation	FR-003	TC-005, TC-006
BB-4: Removed domain still scanned	FR-004	TC-007

The source screenshots show these defects in To Do. Starter tests do not change their Jira status or prove resolution in a production application. See requirements, SRS, WBS, and starter code status.